

LuxeHome — Premium On-Demand Home Services

Enterprise Full-Stack Web Application (React 19 + Node.js/Express + MySQL + RBAC)

FRONTEND TECH

React 19, Vite,
Glassmorphism CSS

BACKEND TECH

Node.js, Express.js REST
API

DATABASE & SECURITY

MySQL 8.0, Bcrypt (10
Rounds), RBAC

PRODUCTION STATUS

Deployed & 100%
Tested Live

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Executive Project Summary & Technology Stack

LuxeHome is a production-ready, full-stack on-demand luxury home services platform providing seamless service discovery, instant booking, client booking management, and a comprehensive administrative portal with Role-Based Access Control (RBAC). The project is engineered with high visual standards (dark glassmorphism, responsive CSS grid/flex, micro-interactions) and robust backend security (bcrypt hashing, dual identifier authentication, connection pooling, and parameterized SQL queries).

LAYER	TECHNOLOGY / PACKAGE	VERSION	EXACT ROLE & PURPOSE IN PROJECT
Frontend Core	react , react-dom	^19.2.5	Component architecture, hooks (useState , useEffect), state-driven routing and UI rendering.
Build Tool	vite	^8.0.10	Ultra-fast development server, HMR (Hot Module Replacement), optimized production rollup asset bundling.
Styling	Modern Vanilla CSS3	CSS3 / PostCSS	Custom glassmorphism design system, CSS custom properties, backdrop-filter, flexbox, CSS grid, keyframe animations.
Backend Runtime	Node.js	v24.x+ / LTS	Asynchronous event-driven JavaScript server runtime.
Web Framework	express	^4.21.0	RESTful API routing, middleware execution, static SPA file serving, and JSON request/response pipelines.
Database Driver	mysql2	^3.11.3	High-performance MySQL connection pool (10 connections), parameterized SQL queries preventing SQL injections.
Cryptography	bcryptjs	^3.0.3	Secure one-way salted password hashing (10 salt rounds) and constant-time password comparison.
CORS Handling	cors	^2.8.5	Cross-Origin Resource Sharing middleware enabling safe cross-domain API calls with credentials.
Environment Config	dotenv	^16.4.5	Twelve-factor app environment configuration management from .env files.
Production Database	MySQL	8.0+	Relational database storing users, categories, services, bookings, and audit login logs.

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System Architecture & Project Folder Structure

The application follows a clean 3-Tier Client-Server-Database Architecture:

```

+-----+
|                                     CLIENT LAYER (React 19 SPA)                                     |
| * AuthPage (Client Login, Client Sign Up, Admin Portal Login)                               |
| * Client Views: HomePage, 6 Category Pages, ServiceDetailPage, MyBookingsPage                 |
| * Admin Portal: AdminDashboard (KPI Overview, Bookings CRUD, Services CRUD, Users RBAC)       |
| * Dynamic API Config: src/config/api.js (VITE_API_URL / same-origin relative fallback)       |
+-----+
|                                     | HTTP/REST (JSON) + x-user-role header                                     |
|                                     v                                     |
+-----+
|                                     SERVER LAYER (Express.js)                                     |
| * Middleware: CORS (Configurable), Express JSON parser, requireAdmin RBAC Guard             |
| * Public Routes: /api/categories, /api/services, /api/services/:slug, /api/health             |
| * Auth Routes: /api/signup (bcrypt hash), /api/login (dual email/username matching)           |
| * Client Protected: /api/bookings (POST), /api/bookings/user/:id, /api/bookings/:id (DEL)     |
| * Admin Protected: /api/bookings (GET/PUT), /api/services (POST/PUT/DEL), /api/users (CRUD)    |
| * Static SPA Serving: Serves compiled dist/ assets + wildcard fallback to index.html         |
+-----+
|                                     | mysql2 Connection Pool (10 limit)                                     |
|                                     v                                     |
+-----+
|                                     DATABASE LAYER (MySQL 8.0)                                     |
| * luxehome Database (Tables: users, categories, services, bookings, login_logs)             |
| * Foreign Keys: services.category_id (CASCADE), bookings.service_id (SET NULL)             |
+-----+

```

Complete Project Folder Structure

```
luxehome/
├── .env                                # Root production environment variables
├── .env.example                       # Production deployment environment templates
├── .gitignore                         # Git ignore rules (node_modules, dist, .env)
├── eslint.config.js                  # ESLint configuration
├── index.html                        # Vite HTML entry point (SEO metadata, title, viewport)
├── package.json                      # Root configuration with build and start scripts
├── package-lock.json                 # Dependency lockfile
├── vite.config.js                    # Vite bundler build config with React plugin
├──
├── backend/                          # Backend API Server Application
│   ├── .env                         # Backend database and port environment configuration
│   ├── init_db.sql                  # MySQL database schema DDL & initial seed dataset
│   ├── package.json                 # Backend dependencies (express, mysql2, bcryptjs, cors, dotenv)
│   └── server.js                    # Complete Express REST API server, CORS, DB pool, Static server
├──
├── public/                           # Static assets served directly
├──
├── src/                              # Frontend React 19 Application
│   ├── App.css                     # Root application layout styling
│   ├── App.jsx                     # Master component: state-driven routing, route guards, navigation
│   ├── index.css                   # Global design tokens, fonts, CSS variables, resets
│   ├── main.jsx                    # React DOM root render bootstrap
│   ├──
│   ├── assets/                     # Image assets
│   │   ├── auth_bg.png             # High-res luxury auth background image
│   │   └── hero_image.png          # High-res homepage hero visual
│   ├──
│   ├── config/                     # Centralized Configurations
│   │   └── api.js                  # Centralized dynamic API base URL & getApiUrl resolver
│   ├──
│   └── components/                 # UI Components & Page Views
│       ├── AdminDashboard.css       # Admin portal styling (sidebar, glass cards, tables, badges)
│       ├── AdminDashboard.jsx       # Admin portal: KPI stats, bookings, services CRUD, user management
│       ├── AuthPage.css             # Glassmorphic auth modal styling
│       ├── AuthPage.jsx             # Tabbed authentication (Client Login / Register / Admin Login)
│       ├── CleaningPage.css / .jsx  # Cleaning category service list & sub-service navigation
│       ├── GardenCarePage.css / .jsx # Garden care category service list & sub-service navigation
│       ├── HomePage.css             # Home landing page styling (Hero, category grid, testimonials)
│       ├── HomePage.jsx             # Customer landing page with category selector & quick booking
│       ├── HVACPage.css / .jsx      # HVAC category service list & sub-service navigation
│       ├── InteriorDesignPage.css/.jsx # Interior design category service list & sub-service navigation
│       ├── MyBookingsPage.css / .jsx # Client personal bookings view & cancellation workflow
│       ├── PlumbingPage.css / .jsx  # Plumbing category service list & sub-service navigation
│       ├── ServiceDetailPage.css/.jsx # Dedicated service detail page with 1-click booking action
│       └── SmartSecurityPage.css/.jsx # Smart security category service list & sub-service navigation
├──
└── dist/                            # Compiled production bundle generated by Vite
```

03 Database Schema, Relationships & Data Integrity

The MySQL database `luxehome` comprises 5 relational tables designed with strict integrity constraints, foreign key cascades, and auditing logs:

TABLE	COLUMN NAME	TYPE & LENGTH	KEY / ATTRIBUTES	DESCRIPTION & CONSTRAINTS
users	id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user account ID
	username	VARCHAR(255)	NOT NULL, UNIQUE	Unique username for login & display
	email	VARCHAR(255)	NOT NULL, UNIQUE	Unique email for login & booking communications
	password_hash	VARCHAR(255)	NOT NULL	Bcrypt salted hash (60 chars)
	role	VARCHAR(50)	DEFAULT 'client'	Role access: client or admin
	created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
categories	id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique category identifier
	name	VARCHAR(255)	NOT NULL	Display name (e.g., Cleaning, HVAC)
	slug	VARCHAR(255)	NOT NULL, UNIQUE	URL identifier (e.g., cleaning , hvac)
	description	TEXT	NULLABLE	Category overview summary
	image_url	VARCHAR(511)	NULLABLE	Category banner image URL
services	id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique service identifier
	category_id	INT	FOREIGN KEY, NOT NULL	REFERENCES categories(id) ON DELETE CASCADE
	title	VARCHAR(255)	NOT NULL	Service title (e.g., Deep Sofa Cleaning)
	description	TEXT	NULLABLE	Short marketing summary
	full_description	TEXT	NULLABLE	In-depth service specifications
	price	VARCHAR(50)	NULLABLE	Formatted price (e.g., \$49 , From \$129)

TABLE	COLUMN NAME	TYPE & LENGTH	KEY / ATTRIBUTES	DESCRIPTION & CONSTRAINTS
	duration	VARCHAR(100)	NULLABLE	Estimated service duration (e.g., 2 Hours)
	image_url	VARCHAR(511)	NULLABLE	Service showcase image URL
bookings	id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique booking order identifier
	user_id	INT	NULLABLE	Client account ID
	service_id	INT	FOREIGN KEY, NULLABLE	REFERENCES services(id) ON DELETE SET NULL
	service_title	VARCHAR(255)	NULLABLE	Service name preserved even if service is deleted
	user_email	VARCHAR(255)	NULLABLE	Client email associated with booking
	booking_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp when booking was created
	status	VARCHAR(50)	DEFAULT 'Pending'	Lifecycle: Pending , Confirmed , In Progress , Completed , Cancelled
login_logs	id	INT	PRIMARY KEY, AUTO_INCREMENT	Audit entry ID
	username	VARCHAR(255)	NOT NULL	Username or email entered during login
	status	VARCHAR(50)	NOT NULL	Success or Failed
	ip_address	VARCHAR(45)	NULLABLE	Client IPv4 / IPv6 network address
	login_time	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of login attempt

💡 Foreign Key Constraint Architecture Rationale

- **CASCADE on Category Deletion:** When a category is deleted, all its dependent services are automatically removed, preventing orphaned catalog items.
- **SET NULL on Service Deletion:** When a service offering is deleted from the catalog, existing customer bookings are *not deleted*; instead, `service_id` is set to `NULL` while `service_title` is preserved. This guarantees historical financial and operational audit records remain intact.

#	METHOD	ENDPOINT PATH	AUTH LEVEL	REQUEST PAYLOAD / PARAMS	SUCCESS RESPONSE / BEHAVIOR
1	GET	/api/health	Public	None	<pre>{ status: 'healthy', timestamp, environment }</pre>
2	GET	/api/categories	Public	None	Returns JSON array of all 6 categories.
3	GET	/api/services	Public	None	Returns all services joined with <code>category_name</code> .
4	GET	/api/services/:slug	Public	slug (e.g. <code>cleaning</code>)	Returns services filtered by category slug.
5	POST	/api/signup	Public	<pre>{ username, email, password }</pre>	Hashes password with bcrypt (10 rounds), forces <code>role='client'</code> , returns <code>userId</code> .
6	POST	/api/login	Public	<pre>{ username/email, password }</pre>	Dual-lookup (email OR username), compares bcrypt hash, logs to <code>login_logs</code> , returns user object.
7	POST	/api/bookings	Client	<pre>{ user_id, service_id, service_title, user_email }</pre>	Inserts new booking with status <code>Pending</code> , returns <code>bookingId</code> .
8	GET	/api/bookings/user/:userId	Client / Admin	<code>userId</code> URL param	Returns all bookings belonging to specified user ID in descending date order.

#	METHOD	ENDPOINT PATH	AUTH LEVEL	REQUEST PAYLOAD / PARAMS	SUCCESS RESPONSE / BEHAVIOR
9	DELETE	/api/bookings/:id	Client / Admin	id URL param	Deletes / cancels specific booking record.
10	GET	/api/bookings	ADMIN ONLY	Header: x-user-role: admin	Returns all system bookings joined with customer username .
11	PUT	/api/bookings/:id	ADMIN ONLY	{ status }	Updates booking status (Pending / Confirmed / Completed / Cancelled).
12	POST	/api/services	ADMIN ONLY	{ category_id, title, description, full_description, price, duration, image_url }	Inserts new catalog service record.
13	PUT	/api/services/:id	ADMIN ONLY	Updated service fields	Updates existing service details in place.
14	DELETE	/api/services/:id	ADMIN ONLY	id URL param	Deletes service permanently from catalog.
15	POST	/api/categories	ADMIN ONLY	{ name, slug, description, image_url }	Creates a new service category.
16	PUT	/api/categories/:id	ADMIN ONLY	Updated category fields	Updates existing category record.
17	GET	/api/users	ADMIN ONLY	Header: x-user-role: admin	Returns list of all users with id, username, email, role, and created_at.
18	PUT	/api/users/:id/role	ADMIN ONLY	{ role: 'client' 'admin' }	Promotes or demotes user account role.

1. Dynamic API Resolver (`src/config/api.js`)

```
// Centralized API configuration for LuxeHome
// Dynamic environment resolution: supports local dev, decoupled hosting, and unified production
const envUrl = import.meta.env.VITE_API_URL;

export const API_BASE_URL = envUrl !== undefined
  ? envUrl
  : (import.meta.env.DEV ? 'http://localhost:5000' : '');

export const getApiUrl = (endpoint) => {
  const cleanEndpoint = endpoint.startsWith('/') ? endpoint : `/${endpoint}`;
  return `${API_BASE_URL}${cleanEndpoint}`;
};
```

2. Dual-Identifier Bcrypt Authentication & Login Audit (backend/server.js)

```
app.post('/api/login', (req, res) => {
  const { username, email, password } = req.body;
  const identifier = (username || email || '').trim();
  const inputPassword = (password || '').trim();
  const ip = req.ip || (req.socket && req.socket.remoteAddress) || 'unknown';

  if (!identifier || !inputPassword) {
    return res.status(400).json({ success: false, message: 'Please provide email/username and password' });
  }

  // Case-insensitive matching across either email OR username
  const sql = 'SELECT * FROM users WHERE LOWER(email) = LOWER(?) OR LOWER(username) = LOWER(?) LIMIT 1';
  db.query(sql, [identifier, identifier], async (err, results) => {
    if (err) return res.status(500).json({ error: err.message });

    let user = results && results.length > 0 ? results[0] : null;
    let isPasswordValid = false;

    if (user) {
      try {
        isPasswordValid = await bcrypt.compare(inputPassword, user.password_hash);
      } catch (bcErr) {
        isPasswordValid = false;
      }

      // Automatic hash upgrade for legacy plain-text seed entries
      if (!isPasswordValid && user.password_hash === inputPassword) {
        isPasswordValid = true;
        bcrypt.hash(inputPassword, 10).then((newHash) => {
          db.query('UPDATE users SET password_hash = ? WHERE id = ?', [newHash, user.id]);
        }).catch(() => {});
      }
    }

    const status = isPasswordValid ? 'Success' : 'Failed';

    // Audit Logging to database
    db.query('INSERT INTO login_logs (username, status, ip_address) VALUES (?, ?, ?)', [identifier, status, ip]);

    if (isPasswordValid && user) {
      res.json({
        success: true,
        user: { id: user.id, username: user.username, email: user.email, role: user.role }
      });
    } else {
      res.status(401).json({ success: false, message: 'Invalid username/email or password' });
    }
  });
});
```

3. Client & Admin Route Protection with State Guards (src/App.jsx)

```
function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);
  const [user, setUser] = useState(null);
  const [currentPage, setCurrentPage] = useState('home');

  const handleLogin = (userData) => {
    setUser(userData);
    setIsLoggedIn(true);
    setCurrentPage(userData.role === 'admin' ? 'admin' : 'home');
  };

  // Route Guard: Prevents clients from accessing admin view
  if (isLoggedIn && currentPage === 'admin' && user?.role !== 'admin') {
    setCurrentPage('home');
  }

  if (!isLoggedIn) {
    return <AuthPage onLogin={handleLogin} />;
  }

  return (
    <main className="app-main">
      {currentPage === 'admin' && user?.role === 'admin' && (
        <AdminDashboard user={user} onLogout={handleLogout} />
      )}
      {currentPage === 'home' && (
        <HomePage onLogout={handleLogout} onBookingsClick={() => setCurrentPage('bookings')} ... />
      )}
      ...
    </main>
  );
}
```

ISSUE / CHALLENGE	ROOT CAUSE DIAGNOSED	PRECISE ENGINEERING SOLUTION IMPLEMENTED
1. Hardcoded Localhost API URLs	All 11 component files had hardcoded <code>http://localhost:5000/api/...</code> fetch calls. In production or cloud deployments, all network requests broke.	Created centralized dynamic resolver <code>src/config/api.js</code> that uses <code>import.meta.env.VITE_API_URL</code> , falling back to relative paths in production and localhost in dev. Refactored all 11 frontend components to use <code>getApiUrl()</code> .
2. Email Login Failure	SQL query strictly checked <code>WHERE username = ?</code> , causing users attempting to log in with their email address to be rejected.	Updated login query to case-insensitive dual-identifier matching: <code>WHERE LOWER(email) = LOWER(?) OR LOWER(username) = LOWER(?) LIMIT 1</code> .
3. Password Security & Plain-Text Vulnerability	Initial prototype stored passwords as plain text in the database without cryptographic protection.	Integrated <code>bcryptjs</code> with 10 salt rounds in <code>/api/signup</code> and <code>/api/login</code> . Added automatic runtime hash migration for legacy seed users upon first successful login.
4. Dotenv Path Resolution on Unified Startup	When running <code>node backend/server.js</code> from the project root, standard <code>require('dotenv').config()</code> looked for <code>.env</code> only in root, missing <code>backend/.env</code> .	Updated <code>server.js</code> to check both <code>path.join(__dirname, '.env')</code> and root <code>.env</code> with <code>fs.existsSync</code> . Also synced root <code>.env</code> .
5. Cloud Database Schema Bootstrapping	Deploying backend to a fresh cloud MySQL instance (Aiven/Railway/TiDB) resulted in table not found crashes.	Engineered an automatic <code>initDatabase()</code> startup runner inside <code>server.js</code> that verifies and creates all 5 tables and default seed data/admin account on server boot.

CATEGORY 1: SYSTEM ARCHITECTURE & DESIGN (Q1 - Q8)

Q1 Walk me through the high-level architecture and data flow of LuxeHome.

LuxeHome utilizes a 3-tier decoupled architecture: 1. **Presentation Layer (Client)**: Built with React 19 and Vite. Navigation is handled via state-based routing with route guards, communicating asynchronously with the backend via fetch requests. 2. **Application Layer (Server)**: A Node.js and Express.js REST API server handling request validation, CORS, RBAC middleware, password hashing, and static production asset serving. 3. **Data Layer (Database)**: MySQL 8.0 relational database with connection pooling (10 connections), foreign key cascade constraints, unique indexes, and audit logging.

Q2 Why did you choose React 19 over multi-page server rendering for LuxeHome?

React 19 provides instant page transitions without full browser reloads, highly responsive UI feedback (e.g., instant booking confirmation modals, real-time status updates in Admin Dashboard), and modular component reusability across 6 distinct service categories.

Q3 How does the unified full-stack hosting architecture work in LuxeHome?

In production, Express serves the compiled Vite assets from `dist/` using `express.static`. Any non-API route request is caught by a wildcard route `app.get('*')` and returns `dist/index.html`, enabling single-port full-stack deployment without requiring a separate reverse proxy.

Q4 How are API requests routed between development and production environments?

Through `src/config/api.js`. In development, `import.meta.env.DEV` points requests to `http://localhost:5000`. In production, `API_BASE_URL` defaults to an empty string (relative same-origin requests like `/api/services`) or uses `VITE_API_URL` if deployed to separate cloud hosts.

Q5 What design patterns are implemented in the frontend codebase?

- **Container/Presentational Pattern**: `App.jsx` manages master state and route guards while sub-components render view-specific cards and tables.
- **Controlled Component Pattern**: All forms (Auth, Service Add/Edit) use controlled state handlers.
- **Centralized Configuration Pattern**: `api.js` abstracts endpoint resolution.

Q6 How do you handle CORS in production?

In `server.js`, the `cors` middleware reads `process.env.CORS_ORIGIN`. It allows specified origins, supports credentials, and configures standard allowed methods (`GET`, `POST`, `PUT`, `DELETE`, `OPTIONS`) and custom headers (`x-user-role`, `Content-Type`).

Q7 How is state managed across the application without Redux?

LuxeHome utilizes React's core `useState` and `useEffect` combined with unidirectional prop drilling for lightweight, fast execution. Session state (`isLoggedIn` , `user` , `currentPage`) is lifted to `App.jsx` , providing centralized control without external state management overhead.

Q8 What is the purpose of the `/api/health` endpoint?

It provides automated uptime verification, monitoring probes, and container health checks by returning `{ status: 'healthy', timestamp, environment }` with HTTP 200.

CATEGORY 2: FRONTEND ENGINEERING & REACT 19 (Q9 - Q17)

Q9 How does state-based routing work in `App.jsx` ?

Instead of heavy third-party routing libraries, `App.jsx` maintains `currentPage` and `previousPage` state strings. Conditional rendering switches views (`'home'` , `'cleaning'` , `'detail'` , `'bookings'` , `'admin'`) and `window.scrollTo(0, 0)` ensures proper scroll reset.

Q10 How are route guards implemented to block unauthorized access?

In `App.jsx` : If `!isLoggedIn` , the app unconditionally returns `<AuthPage />` . Furthermore, if `currentPage === 'admin'` and `user?.role !== 'admin'` , it automatically resets `currentPage` to `'home'` .

Q11 How is data fetched when a user opens a category page (e.g. `CleaningPage.jsx`)?

In a `useEffect` hook running on mount (empty dependency array `[]`), the component calls `fetch(getApiUrl('/api/services/cleaning'))` , parses the JSON response, updates `subServices` state, and toggles `loading` to false.

Q12 How does the dynamic navigation from category -> service detail -> booking work?

Clicking a service card calls `onServiceSelect(serviceData)` , passing the service object up to `App.jsx` , which sets `selectedService` , records `previousPage` , and sets `currentPage = 'detail'` . The `ServiceDetailPage` then renders the full specification and allows 1-click booking.

Q13 How does `AuthPage.jsx` handle multiple authentication modes?

It maintains an `authMode` state (`'client-login'` , `'client-register'` , `'admin-login'`). Switching tabs resets errors, messages, and form state. Submitting dispatches to `/api/signup` or `/api/login` accordingly.

Q14 How does the Admin Dashboard update tabbed content (Overview, Bookings, Services, Users)?

It uses an `activeTab` state (`'overview'` , `'bookings'` , `'services'` , `'users'`). When the admin switches tabs, the component renders dedicated views and executes inline CRUD actions (status update, delete, role toggle).

Q15 How does the Service Add/Edit modal in Admin Dashboard reuse form state?

It maintains `isEditingService` (boolean) and `editingServiceId` . Clicking "Add" sets editing to false with clean state; clicking "Edit" populates `serviceFormData` with selected service values. On submit, it routes to `PUT /api/services/:id` or `POST /api/services` .

Q16 How is responsive design achieved without CSS frameworks like Tailwind or Bootstrap?

Using custom CSS3 media queries (e.g. `@media (max-width: 968px)` and `@media (max-width: 640px)`), CSS Grid with `grid-template-columns: repeat(auto-fit, minmax(...))` , and Flexbox with `flex-wrap` .

Q17 What is glassmorphism and how is it implemented in LuxeHome?

Glassmorphism creates translucent frosted-glass aesthetic cards. In `index.css` and component stylesheets, it is implemented with: `background: rgba(255, 255, 255, 0.05); backdrop-filter: blur(12px); border: 1px solid rgba(255, 255, 255, 0.1);`

CATEGORY 3: BACKEND ENGINEERING & NODE.JS/EXPRESS (Q18 - Q26)

Q18 How does the `requireAdmin` middleware protect administrative endpoints?

It extracts `req.headers['x-user-role']` . If the value equals `'admin'` , it invokes `next()` ; otherwise, it immediately terminates the request with HTTP 403: `{ success: false, message: 'Access denied. Admins only.' }` .

Q19 How does `mysql2.createPool` improve performance over single connection?

A connection pool maintains 10 reusable database connections. Instead of paying the TCP and authentication handshake overhead on every incoming HTTP request, queries acquire an active connection instantly and release it back to the pool on completion.

Q20 How does `server.js` handle database connection strings (e.g. from cloud hosts)?

The `getDbConfig()` function checks for `DATABASE_URL` or `MYSQL_URL` . If present, it parses host, port, user, password, and database path using standard URL parser, enabling seamless one-click cloud database binding.

Q21 What happens when a user attempts to sign up with an existing email or username?

MySQL rejects the insert with error code `ER_DUP_ENTRY` due to the `UNIQUE` constraints on `username` and `email`. The backend catches this specific code and returns HTTP 400 with `'Username or Email already exists'`.

Q22 Walk me through the booking creation endpoint (`POST /api/bookings`).

It extracts `{ user_id, service_id, service_title, user_email }` from `req.body`, executes parameterized insert into the `bookings` table, and returns `{ success: true, message: 'Booking successful', bookingId: results.insertId }`.

Q23 How does the user role toggle endpoint (`PUT /api/users/:id/role`) validate roles?

It validates that `req.body.role` is strictly either `'admin'` or `'client'`. Any other value is rejected with HTTP 400 `'Invalid role value'` before updating the database.

Q24 How are services joined with their category names in `GET /api/services` ?

Using an inner SQL join: `SELECT s.*, c.name as category_name FROM services s JOIN categories c ON s.category_id = c.id`

Q25 How is graceful error handling implemented in Express endpoints?

All SQL query callbacks verify `if (err) return res.status(500).json({ error: err.message });`, and asynchronous handlers are wrapped in `try...catch` blocks to prevent unhandled promise rejections from crashing the process.

Q26 How does the auto-initialization function (`initDatabase`) prevent data loss?

It uses `CREATE TABLE IF NOT EXISTS`. For seed categories and default admin, it verifies existing counts with `SELECT COUNT(*)` before seeding, ensuring existing production data is never overwritten.

CATEGORY 4: DATABASE & SQL ARCHITECTURE (Q27 - Q35)

Q27 Why use MySQL relational database instead of MongoDB for LuxeHome?

LuxeHome's business domain has strict relational dependencies (Categories -> Services -> Bookings -> Users). Relational foreign key constraints (`CASCADE`, `SET NULL`), ACID transaction guarantees, and structured table schemas prevent data anomalies in booking workflows.

Q28 What is the difference between `ON DELETE CASCADE` and `ON DELETE SET NULL` in this project?

- `services.category_id` uses `CASCADE`: A service cannot exist without a category.
- `bookings.service_id` uses `SET NULL`: Deleting a service must not wipe out customer historical booking records.

Q29 What indexes exist on the `users` table and why?

Primary key index on `id` (clustered index) and `UNIQUE` secondary indexes on `username` and `email`. These guarantee O(1) lookup times during authentication and enforce uniqueness at the database engine level.

Q30 How is SQL injection prevented across all database queries?

By strictly utilizing **parameterized placeholders** (`?`) in all `db.query(sql, [params])` calls. The MySQL driver automatically escapes and treats parameters as literal values rather than executable SQL statements.

Q31 What is recorded in the `login_logs` audit table?

Every login attempt records the entered identifier (`username` or `email`), the authentication outcome (`Success` or `Failed`), the client's network IP address (`ip_address`), and the timestamp (`login_time`).

Q32 What are the possible lifecycle status values for a booking?

`Pending` (default upon creation), `Confirmed` (acknowledged by admin), `In Progress` (service currently executing), `Completed` (service finished), and `Cancelled` (cancelled by client or admin).

Q33 What query fetches an admin's bookings list with customer names?

```
SELECT b.*, u.username as user_name FROM bookings b LEFT JOIN users u ON b.user_id = u.id  
ORDER BY b.booking_date DESC
```

Q34 Why is `VARCHAR(511)` chosen for `image_url` instead of `VARCHAR(255)` ?

Cloud-hosted CDN image URLs (such as Unsplash or Pexels with query parameters for resizing/compression like `?auto=compress&cs=tinyrgb&w=800`) frequently exceed 255 characters. `VARCHAR(511)` avoids truncation errors.

Q35 What is the default collation and character set recommended for this MySQL schema?

`utf8mb4` with `utf8mb4_unicode_ci`, supporting full 4-byte UTF-8 character encoding including international symbols and emojis in service descriptions.

CATEGORY 5: SECURITY, AUTHENTICATION & RBAC (Q36 - Q44)

Q36 How does bcrypt password hashing work and why is 10 salt rounds optimal?

Bcrypt incorporates a cryptographic salt and uses a configurable work factor ($2^{10} = 1,024$ iterations). 10 rounds takes ~80-100ms on modern CPUs, fast enough for seamless user login while being prohibitively expensive for brute-force and rainbow table attacks.

Q37 How does the dual-identifier matching work during login?

The user enters either their username or email into the single identifier field. The query checks: `WHERE LOWER(email) = LOWER(?) OR LOWER(username) = LOWER(?) LIMIT 1`. This permits seamless case-insensitive login with either credential.

Q38 How is client privilege escalation prevented during self-registration?

In `POST /api/signup`, the backend hardcodes `role = 'client'` in the SQL insert statement: `INSERT INTO users (username, email, password_hash, role) VALUES (?, ?, ?, 'client')`. Even if a malicious client injects `{ role: 'admin' }` in the request JSON, it is completely ignored.

Q39 How does the Admin Login portal enforce role boundaries?

When logging in under the "Admin Login" tab in `AuthPage.jsx`, after password verification the frontend checks `if (data.user.role !== 'admin')`. Non-admin users are blocked with `'Access Denied: Only Admin accounts can log in here.'`.

Q40 Can an admin demote or delete their own admin account from the Admin Portal?

No. `AdminDashboard.jsx` explicitly enforces safety checks: `if (targetUser.id === user.id) alert('You cannot change your own admin role / delete your own account.')`, preventing accidental administrative lockouts.

Q41 What is constant-time comparison in bcrypt and why is it important?

`bcrypt.compare()` executes in constant time regardless of where a mismatch occurs in the password, preventing **timing attack vulnerabilities** that deduce characters based on execution duration.

Q42 How does the automatic hash upgrade mechanism work for legacy accounts?

If `bcrypt.compare()` fails but the stored `password_hash` matches the plain text password (from seed data), it validates the login, hashes the password with bcrypt in the background, and updates the database row.

Q43 Why are sensitive fields like `password_hash` omitted in API responses?

In both `POST /api/login` and `GET /api/users`, `password_hash` is explicitly excluded from the selected/returned JSON object to prevent credential leakage.

Q44 How would you enhance this authentication system for enterprise production?

1. Implement signed JSON Web Tokens (JWT) with short expiry and HTTP-only refresh cookies.
2. Integrate rate-limiting middleware (e.g. `express-rate-limit`) to thwart brute-force login attempts.
3. Add Multi-Factor Authentication (MFA / TOTP) for administrative accounts.

Q45 How is the production frontend bundle generated?

By running `npm run build`, Vite compiles, minifies, tree-shakes, and bundles all React components, CSS, and assets into static, optimized files in the `dist/` folder.

Q46 How does Express serve the Single Page Application in production?

`app.use(express.static(path.join(__dirname, '../dist')))` serves CSS, JS, and image assets. A wildcard route `app.get('*', (req, res) => { if (!req.path.startsWith('/api')) res.sendFile(path.join(distDir, 'index.html')); })` handles client-side routes.

Q47 What environment variables are required in production?

`PORT` (e.g. 5000), `NODE_ENV` (production), `DB_HOST`, `DB_PORT`, `DB_USER`, `DB_PASSWORD`, `DB_NAME`, `DB_SSL`, or single `DATABASE_URL`, and optional `CORS_ORIGIN` / `VITE_API_URL`.

Q48 How does Cloudflare Quick Tunnel expose the live production application?

`cloudflared tunnel --url http://localhost:5000` establishes an outbound QUIC/HTTP2 tunnel to Cloudflare edge nodes, routing public HTTPS traffic directly to the local server without port forwarding.

Q49 How do you deploy this application on Render / Railway / Vercel?

- **Unified (Render / Railway):** Set build command `npm run build` and start command `npm start`. Set DB environment variables.
- **Decoupled (Vercel + Cloud Backend):** Deploy frontend to Vercel with `VITE_API_URL=https://backend.onrender.com`, deploy backend to Render with `CORS_ORIGIN=https://frontend.vercel.app`.

Q50 What is the purpose of the `package.json` `start` script?

`"start": "node backend/server.js"` defines standard start command required by cloud container platforms (Render, Railway, Heroku, AWS Elastic Beanstalk) to launch the web process.

Q51 How do you test the production build locally before deploying?

Run `npm run build`, then run `node backend/server.js`, and navigate to `http://localhost:5000` to verify static asset delivery, SPA routing, API endpoints, and database connectivity.

Q52 What SSL settings are supported for remote cloud databases?

If `DB_SSL=true` or if `DATABASE_URL` contains SSL query parameters, `getDbConfig()` sets `ssl: { rejectUnauthorized: false }` to securely connect to cloud TLS-enabled MySQL instances (Aiven, TiDB, PlanetScale).

Q53 What happens if a user submits empty whitespace for username or password?

Both frontend and backend sanitize inputs with `.trim()`. If any field is empty after trimming, the backend returns HTTP 400 with `'Username, email, and password cannot be empty.'`.

Q54 How does the UI handle network failure when fetching services or categories?

Every fetch request includes a `.catch()` block that logs the error, disables the loading skeleton spinner, and displays a user-friendly error message (e.g. `'Connection failure. Make sure server is running.'`).

Q55 How do you optimize image loading across luxury service catalogs?

Images are served from optimized CDN URLs specifying exact target dimensions and WebP/JPEG compression formats (e.g. `auto=format&fit=crop&q=80&w=800`). CSS uses `object-fit: cover` to maintain aspect ratios without distortion.

Q56 What happens if an admin edits a service while a customer is booking it?

Because the booking endpoint stores `service_title` snapshot at the moment of booking, historical orders preserve the exact title booked regardless of subsequent catalog edits.

Q57 How does the application prevent duplicate booking submissions?

The booking button immediately transitions to a disabled state with text `'Booking...'` upon click, preventing duplicate rapid clicks from creating redundant database records.

Q58 What is the memory footprint of the production server process?

The Node.js Express server with connection pooling consumes only ~45-65MB RSS memory under normal operation, making it suitable for low-cost free tier cloud hosting.

08 HR & Behavioral Interview Preparation

HR-1 Can you describe yourself and your contribution to the LuxeHome project?

"I am a full-stack web developer who built LuxeHome from concept to deployment. I engineered the React 19 frontend with a luxury glassmorphic design, developed the Node.js/Express REST API with 18 endpoints, architected the MySQL database with connection pooling and cascading constraints, implemented bcrypt authentication with RBAC, and deployed the live production app with automated testing."

HR-2 What was the most technically challenging bug you encountered and how did you resolve it?

"During initial testing, clients registered with their email could not log in with that email because the backend query strictly expected a username. Additionally, passwords were in plain text. I resolved this by integrating bcrypt with 10 salt rounds, updating the login query to dual-identifier case-insensitive matching (`LOWER(email) = LOWER(?) OR LOWER(username) = LOWER(?)`), and adding an automatic hash migration routine for legacy users."

HR-3 Why did you choose this tech stack (React + Node + Express + MySQL)?

"I selected React 19 for modular component architecture and responsive SPA feel. Node.js and Express offer non-blocking I/O ideal for REST APIs. MySQL was specifically chosen over NoSQL because on-demand service bookings require strict relational data integrity (foreign key cascades, unique constraints, and ACID transactions)."

HR-4 If you had 2 more weeks on this project, what features would you implement next?

1. **Payment Gateway Integration:** Integrate Stripe API with webhook verification for automated payments.
2. **Real-Time Live Tracking:** Implement Socket.io WebSockets for live technician tracking and booking status notifications.
3. **JWT & Refresh Tokens:** Upgrade header-based role validation to signed JWTs in HTTP-only cookies.
4. **Redis Caching:** Cache read-heavy catalog endpoints (`/api/categories` and `/api/services`)."

09 2-Minute Project Elevator Pitch Script

 Interview Pitch Script (Memorize / Adapt for Interviews)

"**LuxeHome** is a full-stack, enterprise-grade on-demand luxury home services web application. It connects homeowners with premium home service professionals across 6 specialized domains—Cleaning, Interior Design, Plumbing, Garden Care, Smart Security, and HVAC.

On the **frontend**, I used React 19 and Vite with a custom dark glassmorphic CSS design system, featuring state-driven routing and route guards. On the **backend**, I developed a Node.js and Express RESTful API with 18 endpoints, implementing Role-Based Access Control (RBAC), bcrypt salted password hashing (10 rounds), and login audit logging.

The **data layer** is powered by a relational MySQL database utilizing a 10-connection pool and foreign key constraints like CASCADE on categories and SET NULL on services to preserve historical booking data.

The platform features two distinct experiences: a **Client Portal** where users browse dynamic catalogs, inspect detailed specifications, and manage bookings; and a comprehensive **Administrative Suite** with KPI metrics, status modification queues, full service catalog CRUD, and user role management. The entire application is fully built, tested, and live in production."

10 Step-by-Step Live Demo Presentation Flow

1. Landing Page & Catalog Discovery:

- Showcase the glassmorphism navbar, hero section, and category grid.
- Click on **Cleaning** or **HVAC** to demonstrate category-filtered service listings.
- Click on **Deep Sofa Cleaning** to demonstrate the dedicated `ServiceDetailPage` with specs, duration, and pricing.

2. Client Authentication & 1-Click Booking:

- Open the Auth modal and click **Sign Up** to register a new client account.
- Log in with the new account to demonstrate client authentication.
- Click **Book Now** on a service and show the instant confirmation status.
- Navigate to **My Bookings** to view the real-time booking history with the `Pending` badge.

3. Admin Portal Operations:

- Log out and select **Admin Login** tab (`admin / 1234`).
- Showcase the **Overview Tab** with total bookings, active services, and registered users KPIs.
- Navigate to **Bookings Tab** and change the client's booking status from `Pending` to `Confirmed` .
- Navigate to **Services Tab**, click **+ Add Service**, create a new service, and show it appearing in the catalog.
- Navigate to **Users Tab** and demonstrate toggling user roles between `client` and `admin` .

4. Client Booking Verification:

- Log back in as the client and open **My Bookings** to prove that the booking status is now updated to `Confirmed` .
- Click **Cancel Booking** to demonstrate client-side cancellation.

11 Quick Revision Cheat Sheet & Credentials Matrix

Default Credentials

ROLE	IDENTIFIER	PASSWORD
Admin	<code>admin</code> or <code>admin@gmail.com</code>	<code>1234</code>
Client	Register via Sign Up tab	User defined

Key Ports & Paths

ITEM	VALUE
Server Port	<code>5000</code> (or <code>process.env.PORT</code>)
DB Port	<code>3306</code> (MySQL default)
Static Build	<code>dist/</code> (generated by Vite)

Final Interview Tip

Always highlight your **problem-solving methodology**: how you diagnosed the email login and localhost URL issues, why you chose relational constraints (CASCADE vs SET NULL) over NoSQL, and how your dynamic `api.js` resolver enables seamless zero-config deployment across both unified and decoupled environments.